

Plasma fatty acid composition and incidence of diabetes in middle-aged adults: the Atherosclerosis Risk in Communities (ARIC) Study.

BACKGROUND: The results of some epidemiologic studies conducted by using questionnaires suggest that dietary fat composition influences diabetes risk. Confirmation of this finding with use of a biomarker is warranted. **OBJECTIVE:** We prospectively investigated the relation of plasma cholesterol ester (CE) and phospholipid (PL) fatty acid composition with the incidence of diabetes mellitus. **DESIGN:** In 2909 adults aged 45-64 y, plasma fatty acid composition was quantified by using gas-liquid chromatography and was expressed as a percentage of total fatty acids. Incident diabetes (n = 252) was identified during 9 y of follow-up. **RESULTS:** After adjustment for age, sex, baseline body mass index, waist-to-hip ratio, alcohol intake, cigarette smoking, physical activity, education, and parental history of diabetes, diabetes incidence was significantly and positively associated with the proportions of total saturated fatty acids in plasma CE and PL. The rate ratios of incident diabetes across quintiles of saturated fatty acids were 1.00, 1.36, 1.16, 1.60, and 2.08 (P = 0.0013) in CE and 1.00, 1.75, 1.87, 2.40, and 3.37 (P < 0.0001) in PL. In CE, the incidence of diabetes was also positively associated with the proportions of palmitic (16:0), palmitoleic (16:1n-7), and dihomo-gamma-linolenic (20:3n-6) acids and inversely associated with the proportion of linoleic acid (18:2n-6). In PL, incident diabetes was positively associated with the proportions of 16:0 and stearic acid (18:0). **CONCLUSIONS:** The proportional saturated fatty acid composition of plasma is positively associated with the development of diabetes. Our findings with the use of this biomarker suggest indirectly that the dietary fat profile, particularly that of saturated fat, may contribute to the etiology of diabetes.